The Credentials Arms Race: Understanding Immigrant Over-Education in the U.S. Labor Market

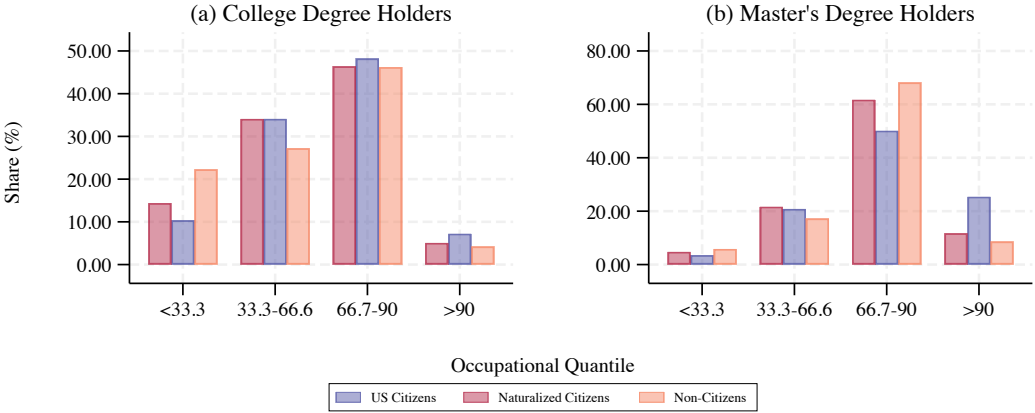
Willy Chen

Empirical Micro Lunch
Feb. 26, 2026

MICHIGAN STATE UNIVERSITY

Motivation

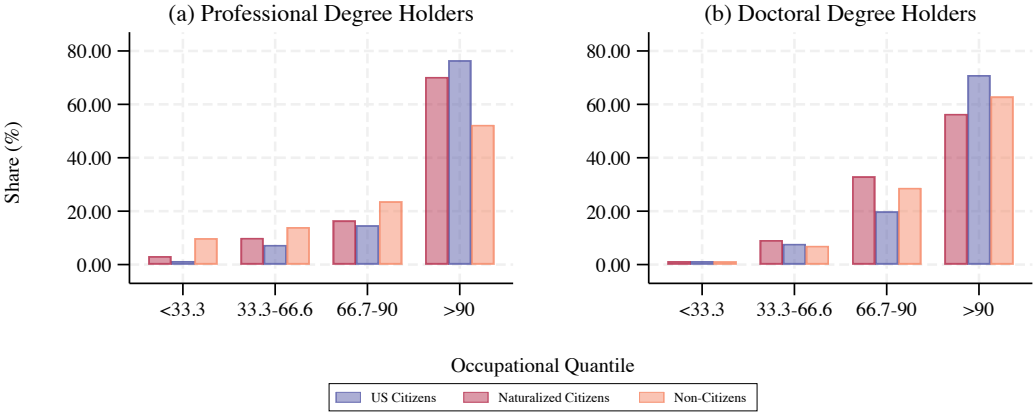
Figure 1: Educational Mismatch of Male Immigrants, 2010-2019 Quartiles



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the U.S. citizen workers' modal degree and their shares of Doctorate/Professional, Master's, College, Associate's, and High School.

Motivation

Figure 2: Educational Mismatch of Male Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the U.S. citizen workers' modal degree and their shares of Doctorate/Professional, Master's, College, Associate's, and High School.

Population Moments

Table 1: Summary Statistics of Workers Age 25-64 in the US with at least High School Education, 2010-2019

	US Citizens	Naturalized citizens	Non-citizens
% of Workforce	85	8.4	6.6
Median Value			
Age	44	46	39
Years in US		24	12
Occ. Quantile	53.5	53.7	40.5
Birthplace Official Language			
% Not English		73.5	74.6
% English		24.7	23.8
% Mixed		1.8	1.6
Highest Degree Obtained			
% High School	50.6	43.1	54.4
% Associate's	10.4	9.5	6
% College	24.9	27.4	21
% Master's	10.1	12.9	12.5
% Professional	2.6	3.9	2.3
% Doctorate	1.4	3.2	3.8
% Single	23.9	15.1	24.4
% Married	59.2	69.9	63.5
Spouse's Citizenship			
US Citizen	95.0	28.0	22.6
Naturalized Citizen	3.1	56.4	16.1
Non-citizen	1.9	15.6	61.3

Related Literature

This paper concerns four strands of literature:

1. Wage Penalty and Over-education (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).

Related Literature

This paper concerns four strands of literature:

1. Wage Penalty and Over-education (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).
2. Language Proficiency and Occupational Assimilation (Beenstock et al., 2010; Chiswick et al., 2009, 2010; Collins et al., 2023; Imai et al., 2019).

Related Literature

This paper concerns four strands of literature:

1. Wage Penalty and Over-education (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).
2. Language Proficiency and Occupational Assimilation (Beenstock et al., 2010; Chiswick et al., 2009, 2010; Collins et al., 2023; Imai et al., 2019).
3. Broader Occupational Mismatch (Cassidy, 2019; Chiswick et al., 2013; Hartog, 2000; Imai et al., 2019; Li et al., 2023; Weber et al., 2024).

Related Literature

This paper concerns four strands of literature:

1. Wage Penalty and Over-education (Cassidy and Gaulke, 2024; Chiswick et al., 2009, 2013; Hartog, 2000; Kiker et al., 1997; Li et al., 2023; Lu et al., 2021; Verdugo et al., 1989; Warman et al., 2015).
2. Language Proficiency and Occupational Assimilation (Beenstock et al., 2010; Chiswick et al., 2009, 2010; Collins et al., 2023; Imai et al., 2019).
3. Broader Occupational Mismatch (Cassidy, 2019; Chiswick et al., 2013; Hartog, 2000; Imai et al., 2019; Li et al., 2023; Weber et al., 2024).
4. Separating equilibria in games with productive signals (Chatterjee et al., 2021; Cho et al., 1987; Kaymak, 2025; Mailath, 1987; Spence, 1978; Tyler et al., 2000).

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.
- The cost of education attainment is $C(\theta, e)$ such that $\frac{\partial}{\partial \theta} C < 0$, $\frac{\partial}{\partial e} C > 0$, $\frac{\partial^2}{\partial e^2} C > 0$, and $\frac{\partial^2}{\partial \theta \partial e} C < 0$.

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.
- The cost of education attainment is $C(\theta, e)$ such that $\frac{\partial}{\partial \theta} C < 0$, $\frac{\partial}{\partial e} C > 0$, $\frac{\partial^2}{\partial e^2} C > 0$, and $\frac{\partial^2}{\partial \theta \partial e} C < 0$.
- The production of a worker in a firm is $F(\theta, e)$ such that $\frac{\partial}{\partial \theta} F, \frac{\partial}{\partial e} F, \frac{\partial^2}{\partial \theta \partial e} F > 0$.

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.
- The cost of education attainment is $C(\theta, e)$ such that $\frac{\partial}{\partial \theta} C < 0$, $\frac{\partial}{\partial e} C > 0$, $\frac{\partial^2}{\partial e^2} C > 0$, and $\frac{\partial^2}{\partial \theta \partial e} C < 0$.
- The production of a worker in a firm is $F(\theta, e)$ such that $\frac{\partial}{\partial \theta} F, \frac{\partial}{\partial e} F, \frac{\partial^2}{\partial \theta \partial e} F > 0$.
- Firms simultaneously offer workers a wage w after observing e.

Model Environment (Modified Spence (1978))

Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

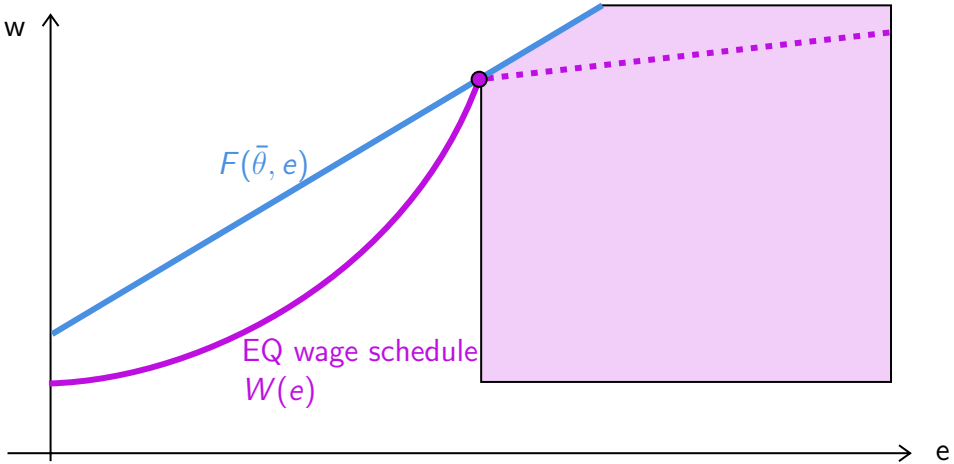
- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.
- The cost of education attainment is $C(\theta, e)$ such that $\frac{\partial}{\partial \theta} C < 0$, $\frac{\partial}{\partial e} C > 0$, $\frac{\partial^2}{\partial e^2} C > 0$, and $\frac{\partial^2}{\partial \theta \partial e} C < 0$.
- The production of a worker in a firm is $F(\theta, e)$ such that $\frac{\partial}{\partial \theta} F, \frac{\partial}{\partial e} F, \frac{\partial^2}{\partial \theta \partial e} F > 0$.
- Firms simultaneously offer workers a wage w after observing e.
- The worker's utility is $U(\theta, e, w) = w - C(\theta, e)$.

Model Environment (Modified Spence (1978))

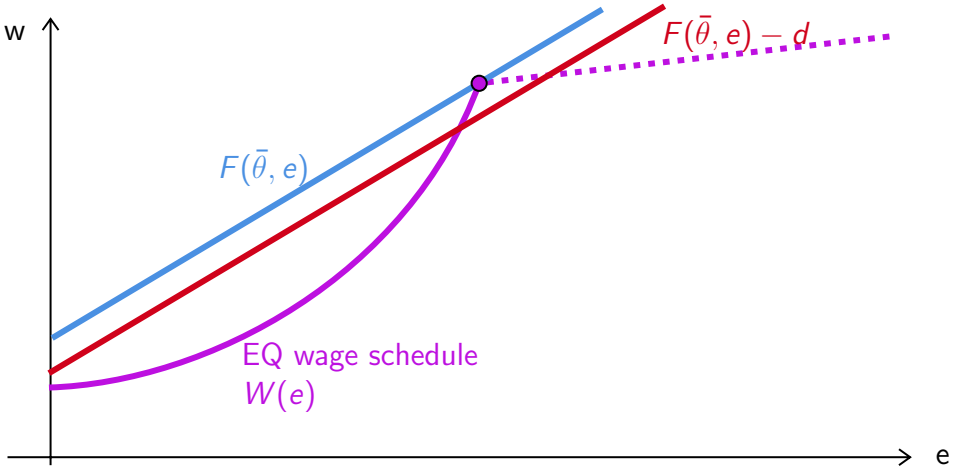
Consider an economy where there are two otherwise identical firms that compete for workers. There is a measure 1 of workers that can be hired.

- Each worker has a private productivity type $\theta \in \Theta \subset \mathbb{R}_+$.
- Each worker chooses an observable education $e \in \mathcal{E} \subset \mathbb{R}_+$.
- The cost of education attainment is $C(\theta, e)$ such that $\frac{\partial}{\partial \theta} C < 0$, $\frac{\partial}{\partial e} C > 0$, $\frac{\partial^2}{\partial e^2} C > 0$, and $\frac{\partial^2}{\partial \theta \partial e} C < 0$.
- The production of a worker in a firm is $F(\theta, e)$ such that $\frac{\partial}{\partial \theta} F, \frac{\partial}{\partial e} F, \frac{\partial^2}{\partial \theta \partial e} F > 0$.
- Firms simultaneously offer workers a wage w after observing e.
- The worker's utility is $U(\theta, e, w) = w - C(\theta, e)$.
- The firm's profit is $\pi(\theta, e, w) = F(\theta, e) - w$.

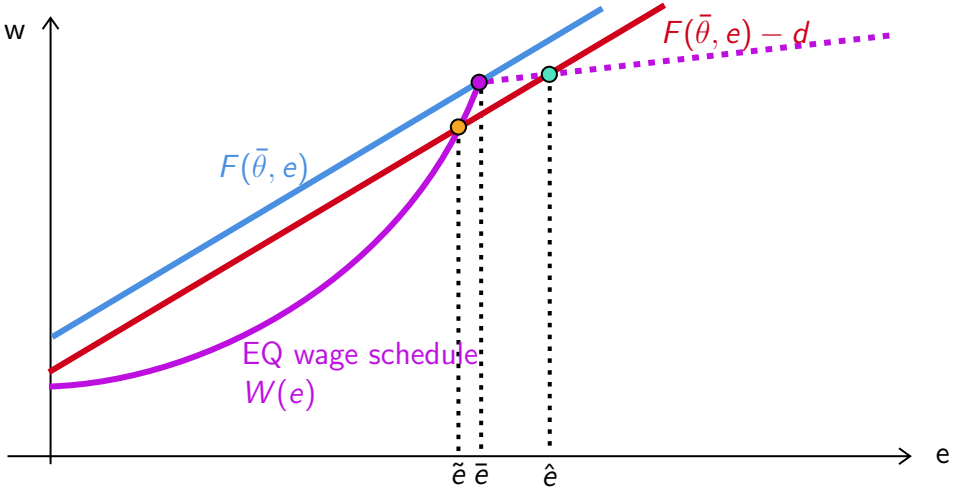
Continuous Types $\Theta = \left[\underline{\theta}, \bar{\theta}\right]$



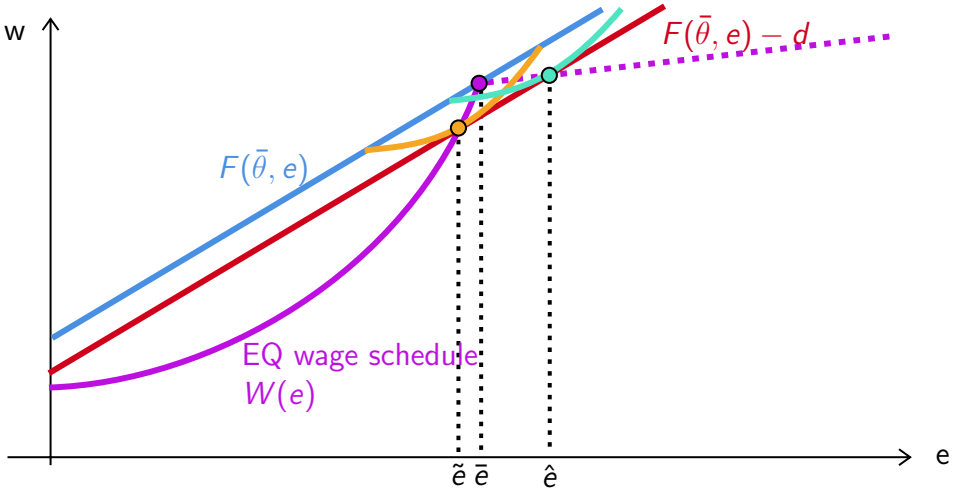
Continuous Types $\Theta = \left[\underline{\theta}, \bar{\theta}\right]$ with New Agents and Costly Hiring



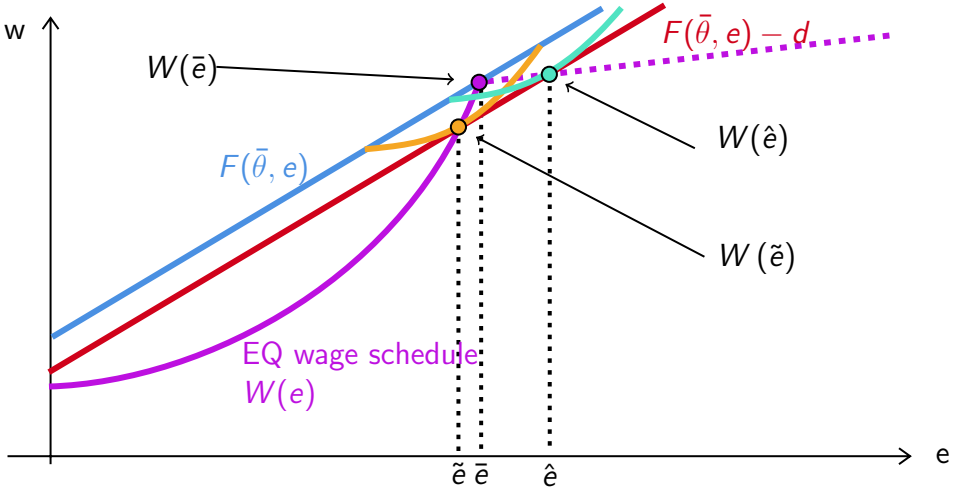
Continuous Types $\Theta = \left[\underline{\theta}, \bar{\theta}\right]$ with New Agents and Costly Hiring



Continuous Types $\Theta = \left[\underline{\theta}, \bar{\theta}\right]$ with New Agents and Costly Hiring



Continuous Types $\Theta = \left[\underline{\theta}, \bar{\theta}\right]$ with New Agents and Costly Hiring



Over-Education vs. Under-Education

The model predicts substantial over-education among highly educated immigrants, but makes no clear prediction for other education levels.

Over-Education vs. Under-Education

The model predicts substantial over-education among highly educated immigrants, but makes no clear prediction for other education levels.

ACS data also shows immigrant *under-education*.

Over-Education vs. Under-Education

The model predicts substantial over-education among highly educated immigrants, but makes no clear prediction for other education levels.

ACS data also shows immigrant *under-education*.

Possible explanations:

- Foreign degrees may be less informative signals $\Rightarrow$ compressed distribution of education attainment and job requirements. In vs. Out

Over-Education vs. Under-Education

The model predicts substantial over-education among highly educated immigrants, but makes no clear prediction for other education levels.

ACS data also shows immigrant *under-education*.

Possible explanations:

- Foreign degrees may be less informative signals $\Rightarrow$ compressed distribution of education attainment and job requirements. In vs. Out
- Some jobs are less desirable, limiting employers' market power and weakening the need for signaling. (Chiswick et al., 2009; Sattinger, 2012)

Over-Education vs. Under-Education

The model predicts substantial over-education among highly educated immigrants, but makes no clear prediction for other education levels.

ACS data also shows immigrant *under-education*.

Possible explanations:

- Foreign degrees may be less informative signals $\Rightarrow$ compressed distribution of education attainment and job requirements. In vs. Out
- Some jobs are less desirable, limiting employers' market power and weakening the need for signaling. (Chiswick et al., 2009; Sattinger, 2012)
  - Some jobs may rely on qualifications other than formal education (Bacon et al., 2020; Chiswick et al., 2008) .

Controlling for Job Characteristics

Controlling for Job Characteristics

1. Principal component analysis (PCA) on O*NET skills and abilities.

Controlling for Job Characteristics

1. Principal component analysis (PCA) on O*NET skills and abilities.
   - Cognitive and Interpersonal Competence Graphs
   - Attention and Coordination
   - Practical Job Capabilities Graphs

Controlling for Job Characteristics

1. Principal component analysis (PCA) on O*NET skills and abilities.
   - Cognitive and Interpersonal Competence Graphs
   - Attention and Coordination
   - Practical Job Capabilities Graphs
2. Importance of Certification vs. Typical On-the-Job Training Graphs

Proxy Wage Schedule

Because realized wages reflect many other factors—bargaining, constraints, unobserved traits—I cannot directly recover the model's true wage schedule from observed wages.

Proxy Wage Schedule

Because realized wages reflect many other factors—bargaining, constraints, unobserved traits—I cannot directly recover the model's true wage schedule from observed wages.

Instead, assuming production and wages increase in both education and type, I proxy the schedule using occupations ranked by their educational composition.

Proxy Wage Schedule

Because realized wages reflect many other factors—bargaining, constraints, unobserved traits—I cannot directly recover the model's true wage schedule from observed wages.

Instead, assuming production and wages increase in both education and type, I proxy the schedule using occupations ranked by their educational composition.

I rank occupations lexicographically by the education distribution of citizen workers and map this ranking into quantiles.

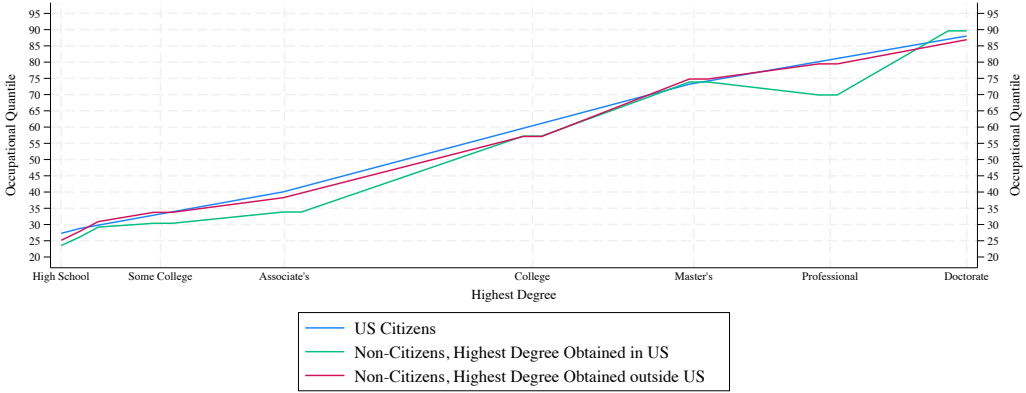
Occupatioal Quantiles

Table 2: Occupational Quantile Example

Occupation	Doc./Pro.	MA	BA	Assoc.	HS	<HS	Occ. Quantile
Surgeon	99.5	0.5	0	0	0	0	100
Postsecondary Teacher	37.8	33.2	19.4	2.4	6.8	0.4	97.1
Sales Managers	1.0	10.8	50.2	7.5	29.5	1.0	61.0
Respiratory Therapists	0.9	3.1	25.0	56.6	14.1	0.3	59.5
Fence Erectors	0	0.7	3.7	4.4	69.4	21.8	3.0
Earth Drillers, not Oil	0	0.2	4.0	6.0	76.3	13.5	2.4

Estimated Proxy Wage Schedule

Figure 3: Estimated "Wage Schedule" by Citizenship, Single Male Age 25-64



Notes: Occupational quantile represents the lexicographic order of the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School. Graph values are smoothed predicted values from separate estimating equations that controls for (1) The interaction of education level and a quadratic specification of experience, (2) Occupational clusters, (3) Year, state, and county FEs, and (4) Quadratic specification of years in the U.S. for immigrants.

Non-citizens have lower experience

New Equilibrium

LinkedIn Profiles from People Data Labs

To study whether employer learning reduces education–occupation gaps—something cross-sectional data (e.g., ACS) cannot address—I construct a panel using scraped LinkedIn profiles from People Data Labs.

Current data include:

- Person record (sex, job, location)
- Job experience (dates, company, title, role)
- Education (dates, school, degree, major)
- Certifications (dates)

LinkedIn Panel

I construct a panel of individuals who attended a U.S. post-secondary institution and held at least one job matched to an O*NET SOC code. The sample covers 1970–2025 and includes ∼175M person-year observations from ∼9.8M individuals, averaging 17.8 years per person (s.d. 9.4).

Distribution of Highest Degree:

- Some College: 67.4%
- Associates: 4%
- Bachelors: 23%
- Masters: 3.9%
- Professional: 1.2%
- PhD & EdD: 0.4%

Population with certifications: 12.1%
Average # of Certifications: 3.5

Average # of jobs after latest edu.: 3.41

- Some College: 3.7
- Assoc.: 3.0
- Bach.: 3.7
- MA.: 3.0
- Prof: 3.4
- EdD: 1.8
- PhD: 3.2

Education Location

Table 3: Distribution of Post-Secondary Degree Location

		Last Post-Sec in US	
		Yes	No
First Post-Sec in US	Yes	73.4%	4.1%
	No	7.9%	5.7%

Education Location

Table 3: Distribution of Post-Secondary Degree Location

		Last Post-Sec in US	
		Yes	No
First Post-Sec in US	Yes	73.4%	4.1%
	No	7.9%	5.7%

Table 4: Average Number of Jobs After Last Degree By Degree Location

Highest Degree	First in US	Last in US	Mean	N	Highest Degree	First in US	Last in US	Mean	N
Bachelors	0	0	4.9	180,468	Professional	0	0	3.8	7,464
	0	1	3.1	249,532		0	1	3.3	23,991
	1	0	3.4	133,257		1	0	3.3	27,142
	1	1	3.7	3,478,333		1	1	3.5	239,546
Masters	0	0	3.8	166,759	PhD	0	0	3.6	26,141
	0	1	3.0	400,409		0	1	3.3	63,111
	1	0	3.0	186,326		1	0	2.9	19,051
	1	1	3.0	1,577,773		1	1	3.2	140,922

Matching LinkedIn Jobs to SOCs

Using O*NET SOC codes linked to LinkedIn job postings from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

Matching LinkedIn Jobs to SOCs

Using O*NET SOC codes linked to LinkedIn job postings from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

This can substantially improves sample retention in the People Data Labs dataset.

Matching LinkedIn Jobs to SOCs

Using O*NET SOC codes linked to LinkedIn job postings from LinkUp, I fine-tune a Llama-3.1 (8B) model to classify each self-reported job title into its corresponding SOC code.

This can substantially improves sample retention in the People Data Labs dataset.

Potential issues: Title inflation leading to observed under-education.

Matching LinkedIn Jobs to SOCs

Table 5: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	**O*NET SOC**	**Corresponding O*NET Title**
director - field support	11-9151.00	Social and Community Service Managers
senior page designer and copy editor	27-3043.00	Writers and Authors
gucci program for scholars	15-1255.00	NA
taylor mattis fellow	51-9195.04	Glass Blowers, Molders, Benders, and Finishers
project management and marketing	15-1299.09	Information Technology Project Managers
solar analyst intern	47-2151.04	NA
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9033.00	Education Administrators, Postsecondary
asset & reliability engineer	17-2112.00	Industrial Engineers
sales / acct manager	11-2022.00	Sales Managers
attest intern	13-1041.04	Government Property Inspectors and Investigators
immersion associate	25-9044.00	Teaching Assistants, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers

Matching LinkedIn Jobs to SOCs

Table 5: Example of Predicted SOC Codes using Preliminarily Fine-Tuned Llama-3.1

LinkedIn Title	**O*NET SOC**	**Corresponding O*NET Title**
director - field support	11-9151.00	Social and Community Service Managers
senior page designer and copy editor	27-3043.00	Writers and Authors
gucci program for scholars	15-1255.00	NA
taylor mattis fellow	51-9195.04	Glass Blowers, Molders, Benders, and Finishers
project management and marketing	15-1299.09	Information Technology Project Managers
solar analyst intern	47-2151.04	NA
international visiting attorney	23-1011.00	Lawyers
director of lower school admission	11-9033.00	Education Administrators, Postsecondary
asset & reliability engineer	17-2112.00	Industrial Engineers
sales / acct manager	11-2022.00	Sales Managers
attest intern	13-1041.04	Government Property Inspectors and Investigators
immersion associate	25-9044.00	Teaching Assistants, Postsecondary
medicaid waiver case manager	11-9151.00	Social and Community Service Managers
網頁工程師	15-1299.08	Computer Systems Engineers/Architects
博士生	25-9044.00	Teaching Assistants, Postsecondary

Future Steps

- Continue to fine-tune SOC code prediction model.

Future Steps

- Continue to fine-tune SOC code prediction model.
- Adopt the model to consider taste-based and statistical discrimination in the US labor market.
  - Using the LinkedIn Panel, I can observe people work outside of the U.S. after their post-secondary degree in the U.S.

Future Steps

- Continue to fine-tune SOC code prediction model.
- Adopt the model to consider taste-based and statistical discrimination in the US labor market.
  - Using the LinkedIn Panel, I can observe people work outside of the U.S. after their post-secondary degree in the U.S.
- Estimate the cost gap d and the proportion of $\bar{\theta}$ productivity immigrants at $\tilde{e}$ and $\hat{e}$.

Future Steps

- Continue to fine-tune SOC code prediction model.
- Adopt the model to consider taste-based and statistical discrimination in the US labor market.
  - Using the LinkedIn Panel, I can observe people work outside of the U.S. after their post-secondary degree in the U.S.
- Estimate the cost gap d and the proportion of $\bar{\theta}$ productivity immigrants at $\tilde{e}$ and $\hat{e}$.
- Counterfactual analysis with increases in d—The new $100,000 H-1B fee.
  - May be able to look at which H-1B jobs on the wage schedule became (un)available after policy takes effect in the 2025-2026 cycle.

Questions?

Thank You!

Bibliography I

Bacon, Kennon and Massimiliano Tani (2020). *Occupational Licensing and Skills Mismatches*. Tech. rep. Working Paper 2020.020. Accessed February 2026. The Center for Growth and Opportunity at Utah State University. URL: https://www.thecgo.org/wp-content/uploads/2020/11/Occupational-Licensing-and-Skills-Mismatches.pdf.

Beenstock, Michael, Barry R Chiswick, and Ari Paltiel (2010). "Testing the immigrant assimilation hypothesis with longitudinal data". In: *Review of Economics of the Household* 8.1, pp. 7–27.

Cassidy, Hugh (2019). "Occupational attainment of natives and immigrants: A cross-cohort analysis". In: *Journal of Human Capital* 13.3, pp. 375–409.

Cassidy, Hugh and Amanda Gaulke (2024). "The increasing penalty to occupation-education mismatch". In: *Economic Inquiry* 62.2, pp. 607–632.

Chatterjee, Somdeep and Jai Kamal (2021). "Long-Term Labor Market Consequences of Costly Signaling: Evidence from a Natural Experiment". In: *Journal of Human Capital* 15.4, pp. 596–628.

Chiswick, Barry R and Paul W Miller (2008). "Why is the payoff to schooling smaller for immigrants?" In: *Labour Economics* 15.6, pp. 1317–1340.

— (2009). "The international transferability of immigrants' human capital". In: *Economics of Education Review* 28.2, pp. 162–169.

Bibliography II

Chiswick, Barry R and Paul W Miller (2010). "Occupational language requirements and the value of English in the US labor market". In: *Journal of Population Economics* 23.1, pp. 353–372.

— (2013). "The impact of surplus skills on earnings: Extending the over-education model to language proficiency". In: *Economics of Education Review* 36, pp. 263–275.

Cho, In-Koo and David M Kreps (1987). "Signaling games and stable equilibria". In: *The Quarterly Journal of Economics* 102.2, pp. 179–221.

Collins, William J and Ariell Zimran (2023). "Working their way up? US immigrants' changing labor market assimilation in the age of mass migration". In: *American Economic Journal: Applied Economics* 15.3, pp. 238–269.

Hartog, Joop (2000). "Over-education and earnings: where are we, where should we go?" In: *Economics of education review* 19.2, pp. 131–147.

Imai, Susumu, Derek Stacey, and Casey Warman (2019). "From engineer to taxi driver? Language proficiency and the occupational skills of immigrants". In: *Canadian Journal of Economics/Revue canadienne d'économique* 52.3, pp. 914–953.

Kaymak, Baris (2025). "Quantifying the signaling role of education". In: *FRB Working Paper Series* 25.02.

Kiker, Billy Frazier, Maria C Santos, and M Mendes De Oliveira (1997). "Overeducation and undereducation: Evidence for Portugal". In: *Economics of Education Review* 16.2, pp. 111–125.

Bibliography III

Li, Xiaoguang and Yao Lu (2023). "Education–occupation mismatch and nativity inequality among highly educated US workers". In: *Demography* 60.1, pp. 201–226.

Lu, Yao and Xiaoguang Li (2021). "Vertical education-occupation mismatch and wage inequality by race/ethnicity and nativity among highly educated US workers". In: *Social Forces* 100.2, pp. 706–737.

Mailath, George J (1987). "Incentive compatibility in signaling games with a continuum of types". In: *Econometrica: Journal of the Econometric Society*, pp. 1349–1365.

Sattinger, Michael (2012). *Assignment Models and Quantitative Mismatches*. Conference Paper. Prepared for the Expert Workshop "Skill Mismatch and Firm Dynamics". University at Albany, Department of Economics. URL: https://melbourneinstitute.unimelb.edu.au/assets/documents/hilda-bibliography/conference-papers-lectures/2012/Sattinger_Assignment_Models_and_Quantitative_Mismatches.pdf.

Spence, Michael (1978). "Job market signaling". In: *Uncertainty in economics*. Elsevier, pp. 281–306.

Tyler, John H, Richard J Murnane, and John B Willett (2000). "Estimating the labor market signaling value of the GED". In: *The Quarterly Journal of Economics* 115.2, pp. 431–468.

Verdugo, Richard R and Naomi Turner Verdugo (1989). "The impact of surplus schooling on earnings: Some additional findings". In: *Journal of human resources*, pp. 629–643.

Bibliography IV

Warman, Casey and Christopher Worswick (2015). "Technological change, occupational tasks and declining immigrant outcomes: Implications for earnings and income inequality in Canada". In: *Canadian Journal of Economics/Revue canadienne d'économique* 48.2, pp. 736–772.

Weber, Rosa, Mathieu Ferry, and Mathieu Ichou (2024). "Which degree for which occupation? Vertical and horizontal mismatch among immigrants, their children, and grandchildren in France". In: *Demography* 61.6, pp. 1923–1948.

Population Moments

Table A6: Summary Statistics of Workers Age 25-64 in the US, 2011-2019

Highest Degree	High School			Associate's			College		
Citizenship	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen
% Workforce	19.9	1.8	2.3	8.8	0.8	0.4	21.2	2.3	1.4
% Single	25	14.8	28.3	21.2	15.1	21.3	26	17.3	21.1
% Married	54.8	67.9	58.3	60.1	67.6	62.5	62	69.9	68.6
Spouse's Citizenship									
% Citizen	59.4	15.2	11.4	63.3	20	20.9	64.1	20.8	16.5
% Nat. Citizen	1.3	38	10.6	1.8	38.6	13	2.3	41.1	10.7
% Non-citizen	1.2	15.4	38.9	1.2	10.4	30.1	1.3	9.1	40.5
Birthplace Official Language									
% Not English		80.1	87.5		74.8	74.5		72.1	64.3
% English		18.4	11.5		23.1	22.7		26.1	33.6
% Mixed		1.4	1.1		2.1	2.9		1.8	2
Median Value									
Age	46	47	39	44	46	41	42	45	39
Years in US		24	14		25	14		23	9
Occ. Quantile	32.5	32.4	22.4	51	47.3	40.5	73	71.9	65.2

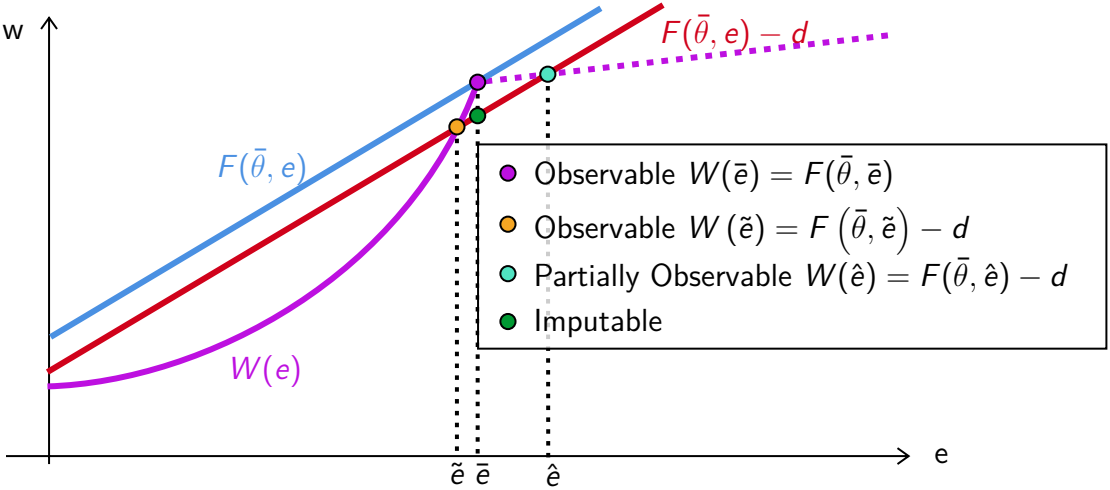
Population Moments

Table A7: Summary Statistics of Workers Age 25-64 in the US, 2010-2019

Highest Degree	Master's			Professional			Doctorate		
Citizenship	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen	Citizen	Nat. Citizen	Non-citizen
% Workforce	8.5	1.1	0.8	2.2	0.3	0.2	1.2	0.3	0.2
% Single	19.8	12.1	20.5	18.2	13.9	19.9	17.3	9.7	18.5
% Married	68.1	76.2	73.6	71.4	76.1	71.5	71.9	81.2	76.1
Spouse's Citizenship									
Citizen	67.8	22.4	13.1	69.9	26	15	68.8	22.5	13.1
Nat. Citizen	2.9	46.3	7.9	4	44.5	9.8	4.5	50.7	6.4
Non-citizen	1.5	8	51.3	1.6	6.5	45.4	2.4	7.4	54.1
Birthplace Official Language									
% Not English		62	50.2		63.8	59.2		68	71.2
% English		35.8	48		34.2	39.4		29.6	27.4
% Mixed		2.3	1.8		2	1.4		2.5	1.4
Median Value									
Age	44	46	37	45	47	39	46	49	39
Years in US		23	8		25	8		24	10
Occ. Quantile	83	80	79	98.9	98.9	95.6	97.1	96.8	96.8

New Separating Equilibrium

Back



General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

Each worker chooses their education according to $\sigma : \Theta \to \mathcal{E}$.
Observing e, the firm updates belief about the worker's type $\widehat{\Theta}(e) \subseteq \Theta$.

General Separating Equilibrium

The general model follows the work of (Mailath, 1987).

Each worker chooses their education according to $\sigma : \Theta \to \mathcal{E}$.
Observing e, the firm updates belief about the worker's type $\widehat{\Theta}(e) \subseteq \Theta$.

In a full **separating** equilibrium, σ is a one-to-one function, i.e.,

$$\widehat{\Theta}(e) = \{\sigma^{-1}(e)\}, \forall e \in \mathcal{E} \tag{1}$$

and $\sigma(\theta)$ must satisfy *strict incentive compatibility*:

$$\{\sigma(\theta)\} = \underset{e}{argmax}\ U(\theta, \sigma^{-1}(e), e), \forall \theta \in \Theta \tag{2}$$

Regularity Conditions

1. (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
2. (Belief Monotonicity) $\dfrac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$.
3. (Type monotonicity) $\dfrac{\partial^2}{\partial \theta \partial e} U\left(\theta, \hat{\theta}, e\right) > 0$
4. ("Strict" Quasi-Concavity) $\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right) = 0$ has a unique solution in e, denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{argmax}\ U\left(\theta, \hat{\theta}, e\right)$, and $\dfrac{\partial^2}{\partial e^2} U\left(\theta, \hat{\theta}, e\right)\Big|_{e=\phi(\theta)} < 0$.
5. (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \ \Rightarrow \ \left|\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)\right| > k.$$

Regularity Conditions

1. (Smoothness) $U(\theta, \hat{\theta}, e)$ is twice differentiable on $\Theta^2 \times \mathcal{E}$.
2. (Belief Monotonicity) $\dfrac{\partial}{\partial \hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$.
3. (Type monotonicity) $\dfrac{\partial^2}{\partial \theta \partial e} U\left(\theta, \hat{\theta}, e\right) > 0$
4. ("Strict" Quasi-Concavity) $\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right) = 0$ has a unique solution in e, denoted $\phi(\theta)$ such that $\phi(\theta) \in \underset{e}{argmax}\ U\left(\theta, \hat{\theta}, e\right)$, and $\dfrac{\partial^2}{\partial e^2} U\left(\theta, \hat{\theta}, e\right)\Big|_{e=\phi(\theta)} < 0$.
5. (Boundedness) $\exists k > 0$ such that

$$\forall (\theta, e) \in \Theta \times \mathcal{E}, \frac{\partial^2}{\partial e^2} U(\theta, \hat{\theta}, e) \geq 0 \ \Rightarrow \ \left|\frac{\partial}{\partial e} U(\theta, \hat{\theta}, e)\right| > k.$$

Additionally, there are two niceness conditions: Initial Value (IV) and Single Crossing (SC).

Characterization of the Solution

Mailath (1987) shows that
If conditions 1-5 are satisfied, then a strictly increasing function σ, continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

1. σ solves $\dfrac{d\sigma}{d\theta} = -\dfrac{\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}$
2. when $\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$, $\dfrac{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}$ is a strictly increasing function of θ.

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

Characterization of the Solution

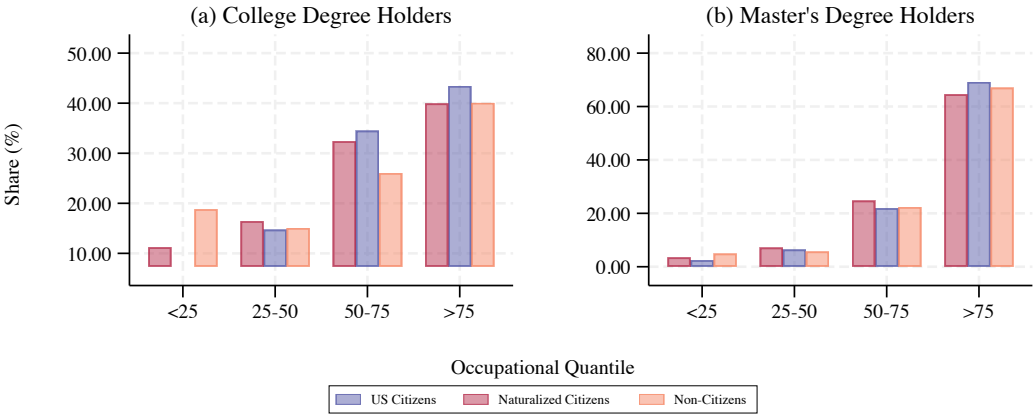
Mailath (1987) shows that
If conditions 1-5 are satisfied, then a strictly increasing function σ, continuous at $\underline{\theta}$, satisfies strict incentive compatibility if and only if:

1. σ solves $\dfrac{d\sigma}{d\theta} = -\dfrac{\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}$
2. when $\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right) > 0$, $\dfrac{\frac{\partial}{\partial e} U\left(\theta, \hat{\theta}, e\right)}{\frac{\partial}{\partial\hat{\theta}} U\left(\theta, \hat{\theta}, e\right)}$ is a strictly increasing function of θ.

Moreover, the strictly increasing σ that satisfies strict incentive compatibility is unique.

In other words, there is an unique full separating equilibrium in the natives' labor market given the assumptions in the basic model environment.

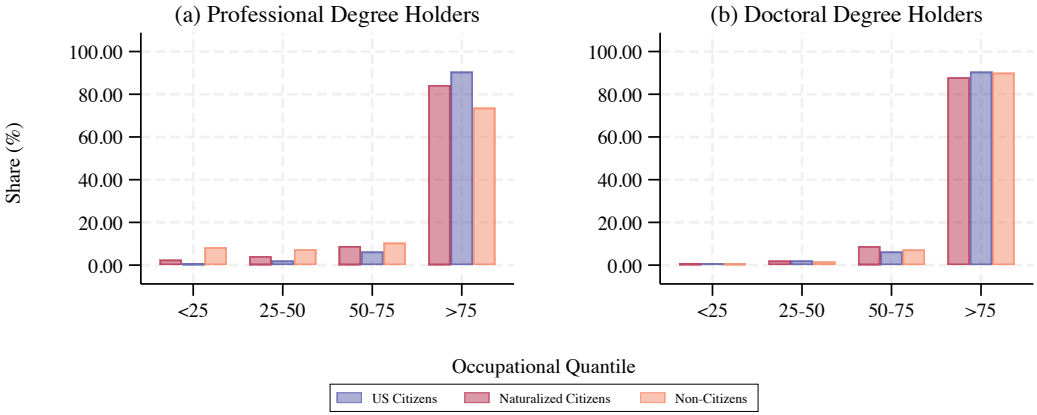
Figure A4: Educational Mismatch of Male Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

Back

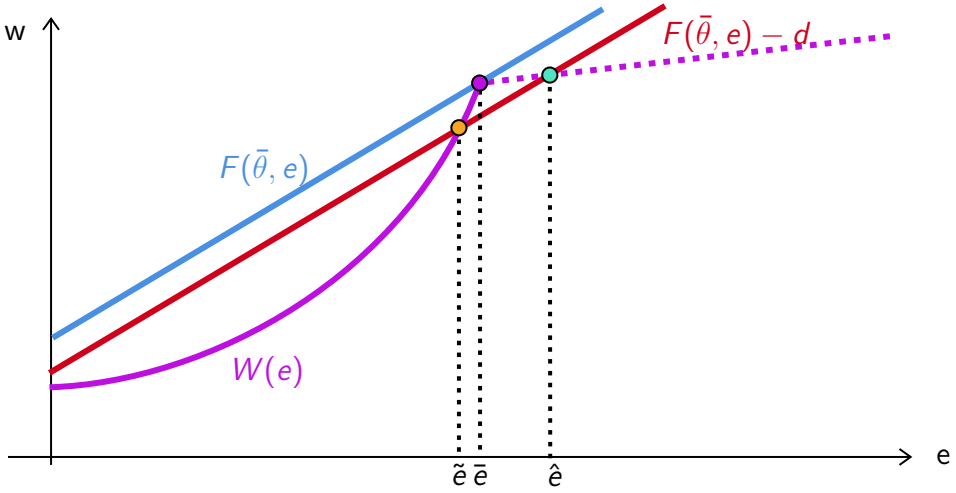
Figure A5: Educational Mismatch of Male Immigrants at High Degree Levels, 2010-2019



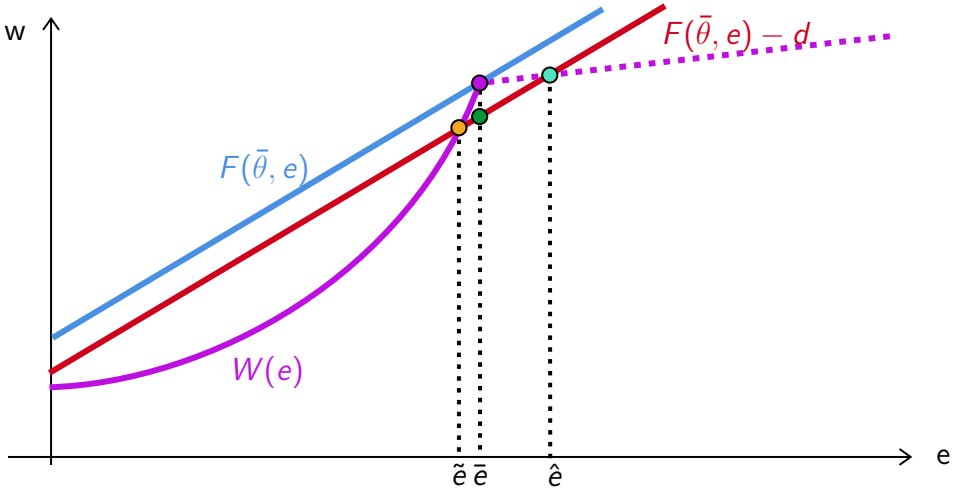
Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

Back

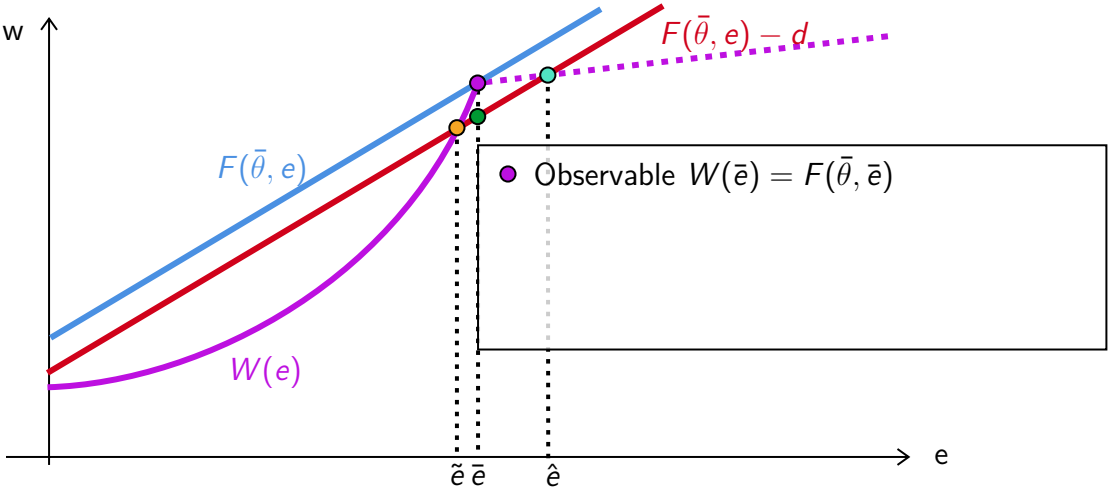
New Separating Equilibrium



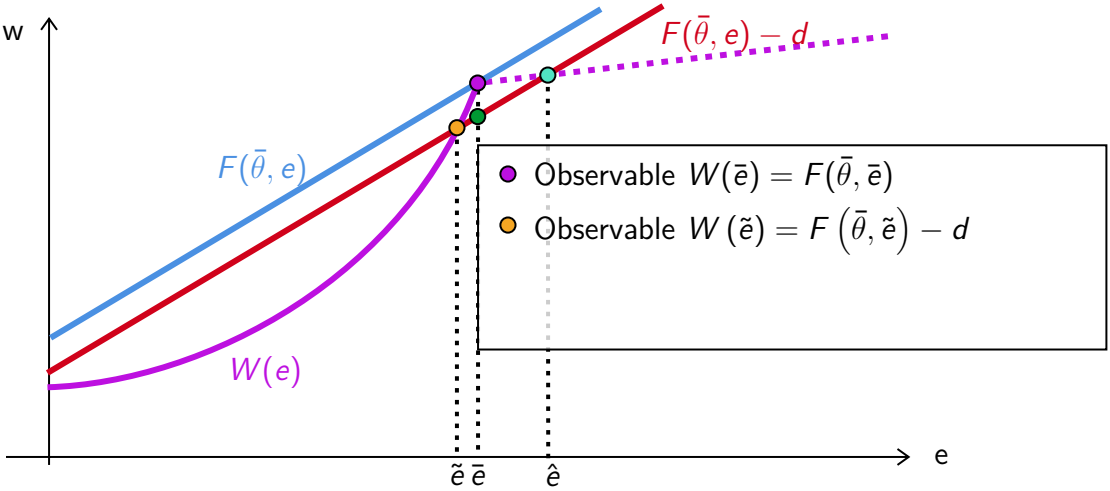
New Separating Equilibrium



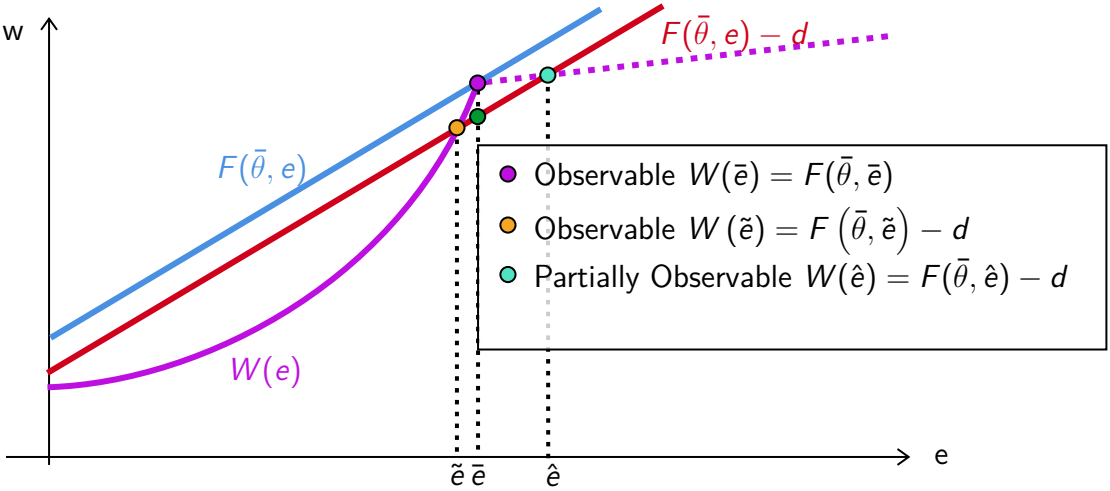
New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium



New Separating Equilibrium

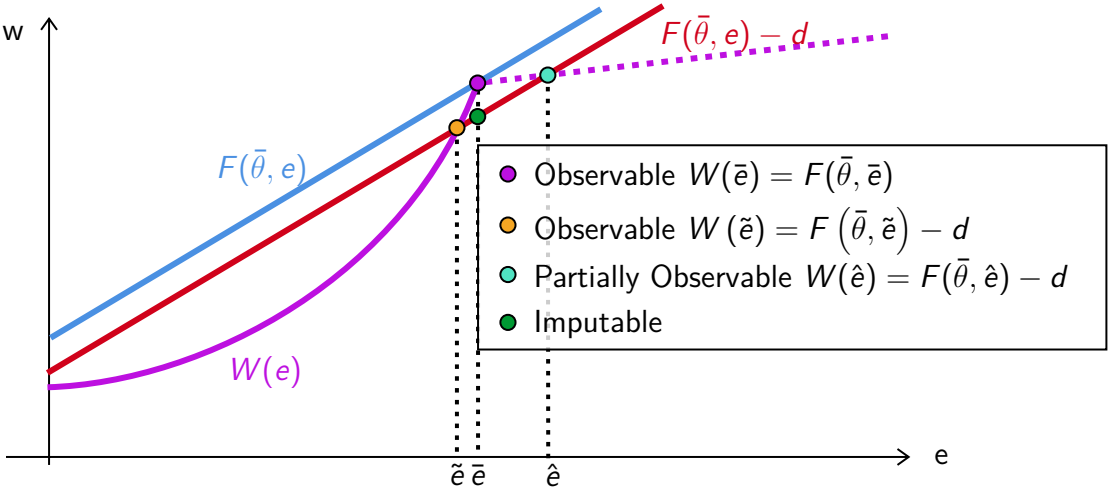
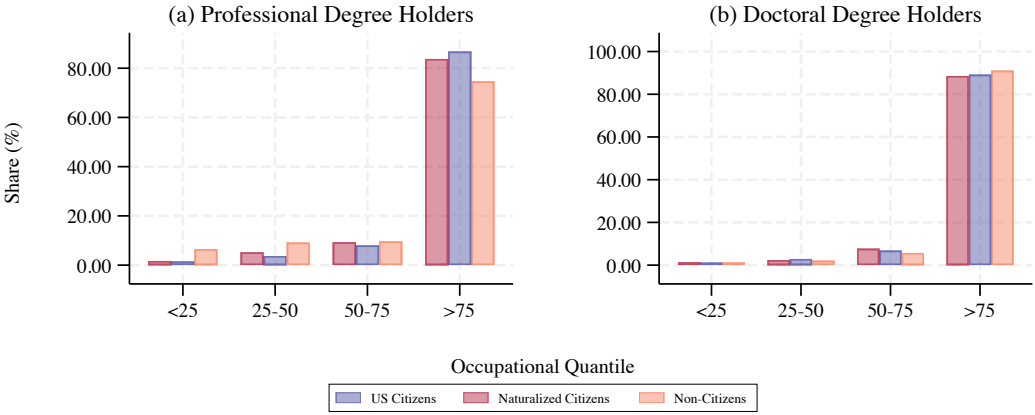


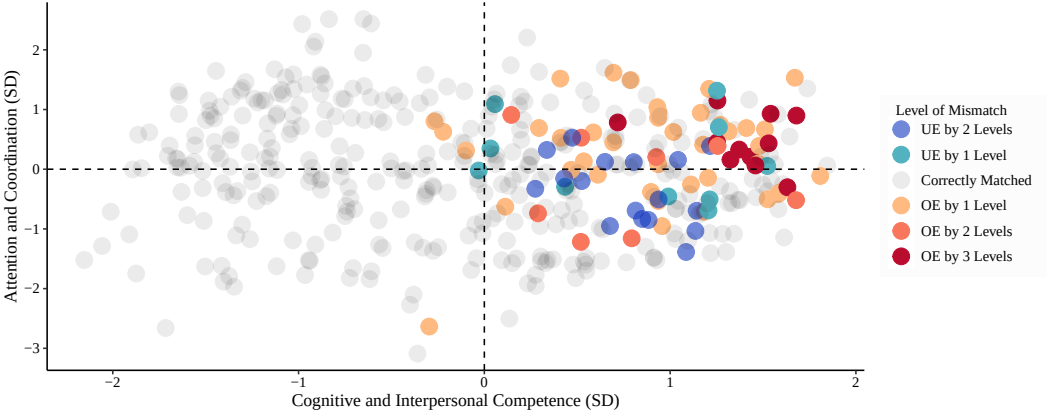
Figure A6: Educational Mismatch of Single Immigrants at High Degree Levels, 2010-2019



Notes: Data from American Community Survey-5% from 2010 to 2019. Occupational quantile represents the lexicographic order of jobs over the shares of U.S. citizen workers' education attainment of Doctorate/Professional, Master's, College, Associate's, and High School.

Back

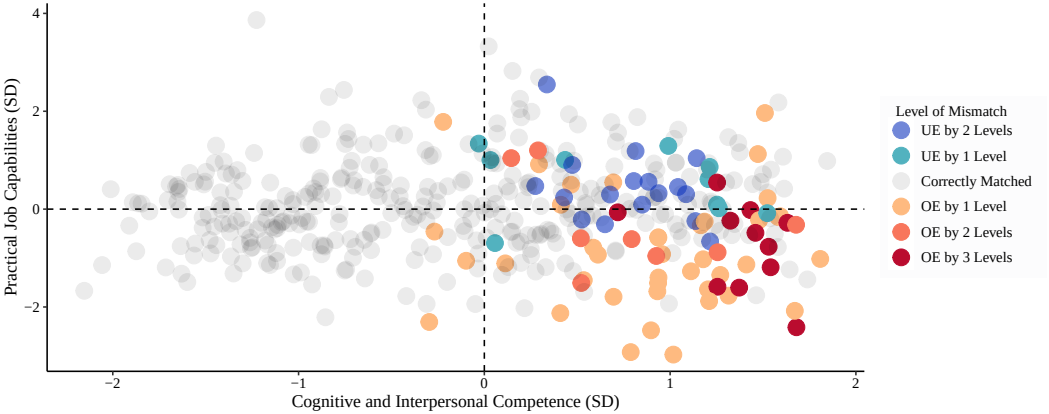
Figure A7: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

Back

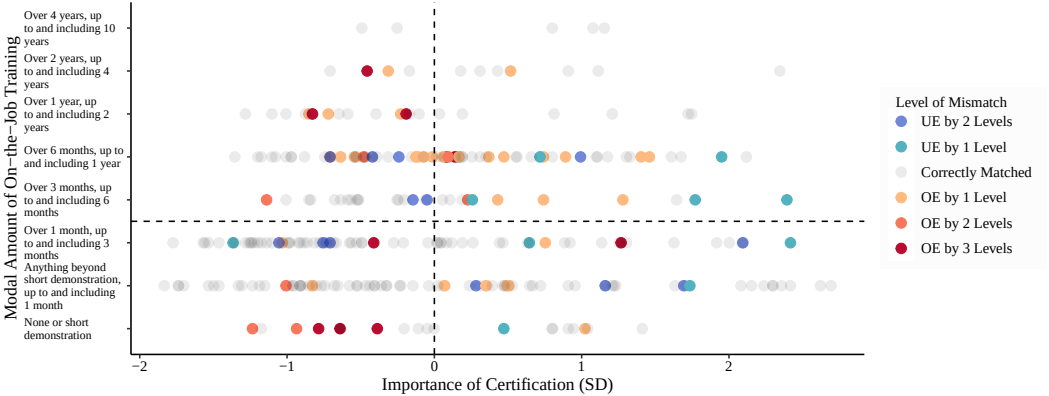
Figure A8: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US



Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School.

Back

Figure A9: Educational Mismatch of Non-Citizens, Highest Degree Obtained in US

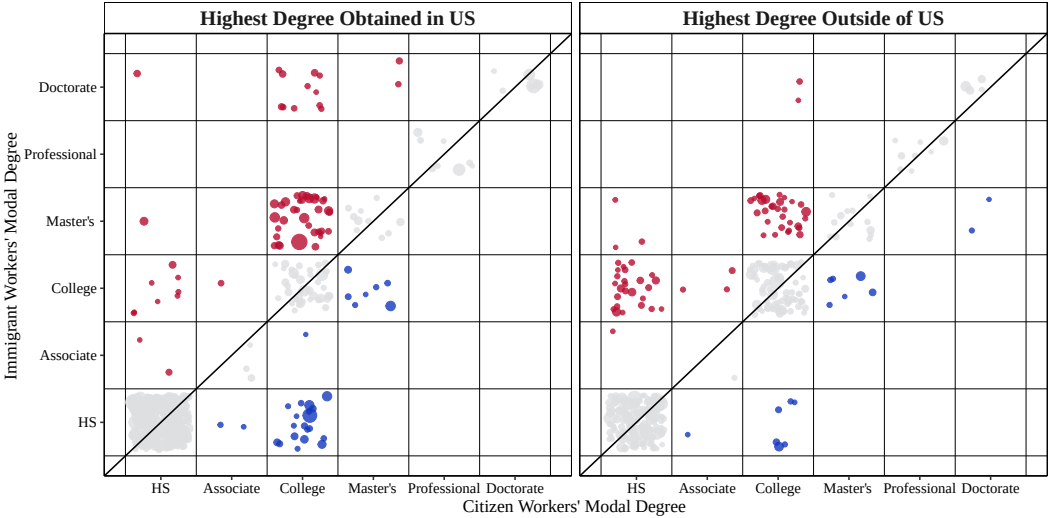


Notes: Level of mismatch is the difference between the modal degree of a non−citizen worker, whose highest degree is obtained inside the United States, and the modal degree of a citizen worker. The degree levels, from highest to lowest, are: Doctorate, Professional, Master's, Bachelor's, Associates, High School, and less than High School. Job characteristics are from the O*NET dataset. Mismatch realizations are from the American Community Survey 2010−2019.

Back

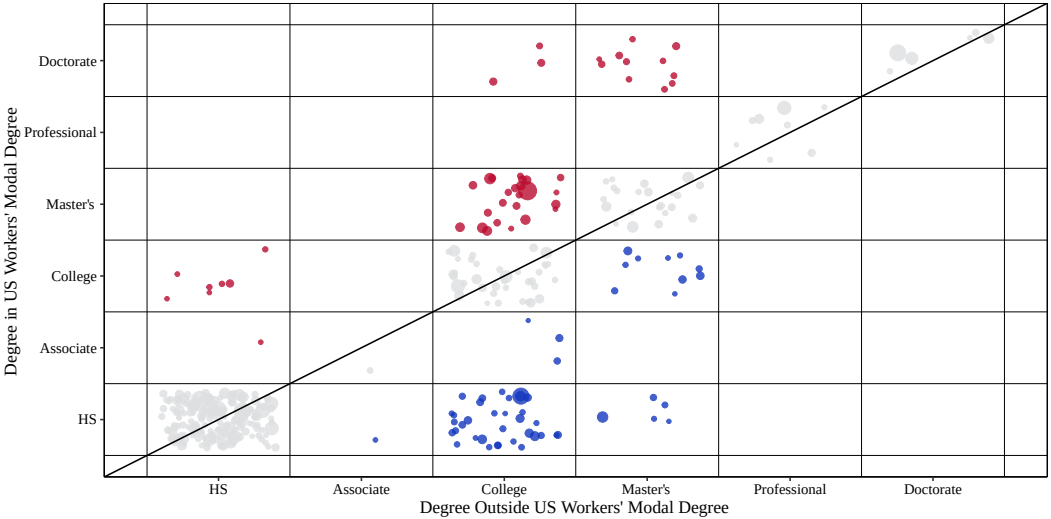
Location of Highest Degree Back

Figure A10: Educational Mismatch of Non-Citizens, 2010-2019



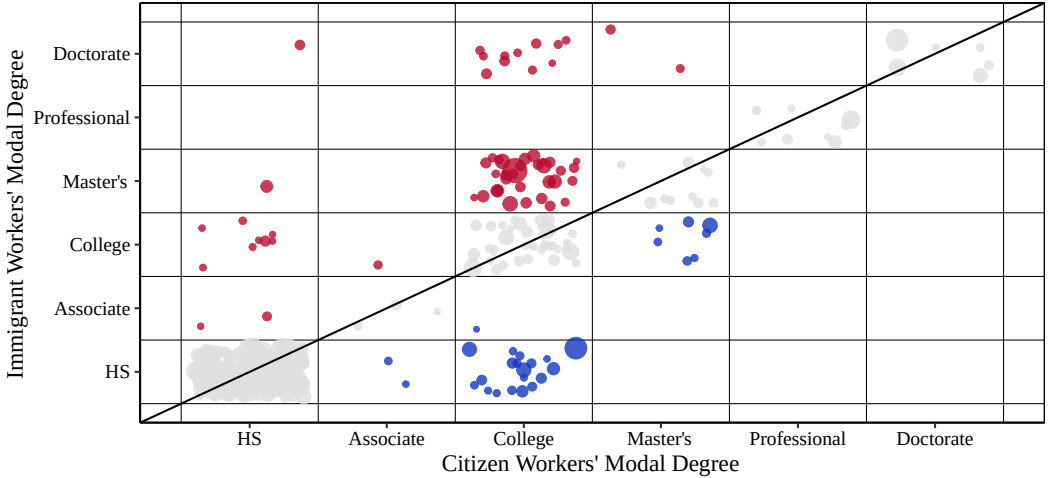
Location of Highest Degree Back

Figure A11: Educational Mismatch of Non-Citizens, 2010-2019



Location of Highest Degree Back

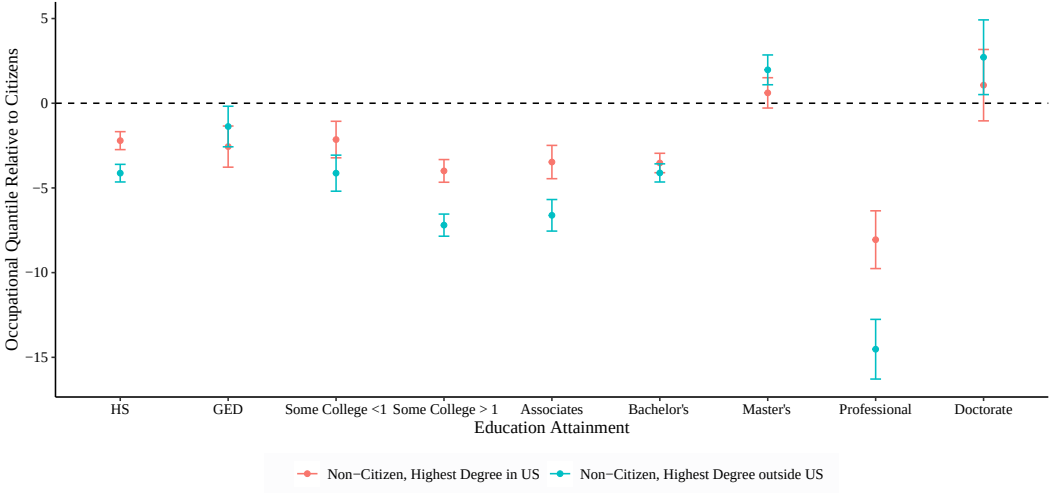
Figure A12: Educational Mismatch of Non-Citizens, Highest Degree in U.S., 2010-2019



Location of Highest Degree Back

Figure A13: Educational Mismatch of Non-Citizens, Highest Degree Outside of U.S., 2010-2019

Figure A14: Estimated Deviations in Mean Occupational Quantile by Education Level for Immigrant Workers, 2010-2019 Back



Notes: 95% confidence intervals are shown in graph.